

SHUBHAM KUMAR SAHOO

Fullstack Engineer | Distributed Systems | Agentic AI Systems | Workflow Automation

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Experience

Hinge Health

Oct 2025 - Mar 2026

Software Engineer II

Bengaluru

- Engineered an RAG system, using **AWS Bedrock Agents** and APIs to leverage historical conversations, for compliance queries, reducing average response time by **80%** (from **3 minutes** to **30 seconds**) and cutting reliance on external coaches for repeated queries (**>50%**).
- Scaled synthetic-data throughput by **1600x** - from **15 samples/day** to **24,000/day** - by building a self-orchestrating generation ecosystem using LangGraph parallelism, LangChain orchestration, async processing, and proprietary LLMs for edge-case coverage and assistant improvement.
- Cut retrieval regressions by **90%** across CI workflows and reduced retrieval latency by **3ms**, by implementing a Corrective RAG subsystem with deterministic seed control, adaptive K-selection, and **750-1000** golden test cases.
- Contributed to an RAG-based internal assistant (**HingeGPT**) ft. multi-source ingestion connectors for GitHub, Jira and Confluence, semantic vector search for context retrieval, improving developer productivity by **30%**.
- Migrated direct user search from PostgreSQL to **OpenSearch**, eliminating **99%** CPU spikes and slashing p95 latency from **>1s** to sub-second performance.
- Automated unit test creation for NestJS/Go via LLMs saving **160 minutes** per class.
- Led the extraction of tightly coupled modules from a **Ruby on Rails** monolith into **5** domain-driven microservices on **NestJS**.

Astria Digital

Feb 2025 - May 2025

Software Engineer

Bengaluru

- Built an event-driven agent communication platform, enabling **50** concurrent agents to handle inbound customer calls with real-time routing and post-call analysis, reducing average handle time by **~30%**.
- Optimized **S3** access for **EKS** workloads by configuring VPC Gateway Endpoints and introducing a **Redis** caching layer for hot objects, eliminating public egress costs and reducing asset fetch latency by **~35%**.
- Improved dashboard UI performance by caching frequently fetched live data, reducing interaction latency by **~28%** and achieving **LCP** under **2.5s**.

Lifesight

Jul 2022 - May 2024

Associate Frontend Engineer

Bengaluru

- Engineered the customer engagement module using Unlayer to enable drag-and-drop email templates and interactive form creation, leading to **1000+** upvotes and **700+** customer signups on **ProductHunt**.
- Built a centralized integrations module UI to unify **10+ marketing platforms** (e.g., Google Ads, Facebook, Shopify).
- Owned major marketing measurement UI modules (Attribution, MMM and Incrementality with Causal Inference).

Technical Skills

Programming & Scripting: Go, JavaScript, TypeScript, Ruby, Python, C++

Frontend: React, React Native, Next.js, Webpack, Rollup, Jest, Redux, SCSS, Tailwind, GSAP, Unlayer, Figma

Backend & Services: Node.js, Express, NestJS, Firebase, Websockets, Distributed Systems, Redis, Kafka, EKS, Opensearch

Data Platform & Engineering: PostgreSQL, MongoDB, AWS, Google Cloud, QDrant

DevOps & Tools: Linux, Docker, Kubernetes, Jenkins, Railway, Expo, Git, Bitbucket, Twilio, Datadog

AI Core: Gen AI, LLMs, Transformer Models, NLP, LLM Fine-Tuning, RAG, Vector Search, Model guardrails, LangChain, LangGraph

Projects

HeroAI | Hinge Health | Hackfest 2026

- Built an autonomous PR auto-healer that triggers when CI fails, analyzes the error logs, and pushes a fix to the branch.

HiRider | React, Typescript, Go, GCP, MongoDB, RabbitMQ, Kubernetes, Stripe

- Built a ride-booking platform using an event-driven architecture with RabbitMQ for asynchronous processing.

Education

PES University, Bengaluru

2018 – 2022

B.Tech in Computer Science and Engineering

Delhi Public School Ruby Park, Kolkata

2018

CBSE (XII)